```
  function brushed() {
     // brushed code
  };
};
```

In particular, we use `d3.time.hours` in the `ticks()` function to create an axis with ticks every two hours. We then use `d3.time.format` in the `tickFormat()` function to show only the hour and whether it's A.M. or P.M.

The `brushed()` function (listing 9.15) that we previously defined in the `createBrush` function gets the current extent of the brush using its built-in function and compares that extent to the two datasets we used for our circle pack chart and bar chart. It then adjusts the size of the bars in the bar chart or the visibility of the circles in the circle pack based on whether the corresponding data falls within the range of the current extent.

Listing 9.15 The `brushed` function

```
function brushed() {
    var e = timeBrush.extent();

       d3.selectAll("circle.pack")
            .filter(function(d){return d.depth == 2})
            .style("display", function (d) {
                 return new Date(d.timestamp) >= e[0] &&
                     new Date(d.timestamp) <= e[1] ? "block" : "none";
            });

       var rightSize = canvasSize("#rightSVG");

       d3.select("#rightSVG")
         .selectAll("rect")
         .attr("x", function(d,i) {return barXScale(d.key) + 5})
         .attr("width", function() {return barXScale.rangeBand() - 5})
         .attr("y", function(d) {return barYScale(filteredLength(d))})
         .style("stroke", "black")
         .attr("height", function(d) {
              return rightSize[1] - barYScale(filteredLength(d));
         });

    function filteredLength(d) {
       var filteredValues = d.values.filter(function (p) {
         return Date(p.timestamp) >= e[0] && new Date(p.timestamp) <= e[1];
       });
       return filteredValues.length;
    };
};
```

Gets the current extent of the brush

Hides any depth 2 circles not tweeted within the current extent

Redraws the bar chart to reflect only tweets within the current extent

This brush allows users to designate a block of time during the day that the tweets are made. It shows only those level 2 circles on the circle pack that correspond to tweets that were made during that time. It also adjusts the bar chart to show totals for the tweets made during that time. Figure 9.9 shows three different brushed regions and the corresponding changes to the circle pack chart and bar chart.